MD. AMIT HASAN

Oceanography Graduate | Participatory Researcher | Field & Data Support | GIS/RS
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Portfolio: amithasanbd.github.io | ResearchGate | LinkedIn



PROFESSIONAL SUMMARY

Detail-oriented Oceanography researcher with 4+ years of combined experience supporting climate, coastal, and environmental research as a Research Associate, Project Coordinator, and Data Analyst. Skilled in GIS/remote sensing (ArcGIS Pro, QGIS, Google Earth Engine), Basic Ocean/numerical modeling (Delft3D, CROCO), statistical and geospatial data analysis (Python, R, SPSS, Power BI), field/survey data collection, and technical reporting. Proven record of coordinating multi-stakeholder research projects, producing decision-support outputs, and co-authoring peer-reviewed publications, with specialized training from NASA (ARSET), UNESCO/IOC, and NOAMI. Currently pursuing an M.Sc. in Oceanography at Bangladesh Maritime University; seeking to bring research coordination, data analysis, and technical reporting expertise to a Research oriented any relevant role.

CORE SKILLS

Remote Sensing & GIS	ArcGIS Pro, QGIS, Google Earth Engine, SeaDAS, geospatial & satellite-derived analysis, field validation
Numerical Modeling	Basic Delft3D, CROCO, OFS configuration, Basic to Advanced XAI modeling
Data & Statistical Analysis	Python, R, SPSS, MS Excel, Power BI, OriginPro, descriptive statistics, data cleaning & validation, dashboards
Research & Technical Writing	Literature review, evidence synthesis, technical/report writing, manuscript support, figure & table preparation
Project & Field Coordination	Project documentation, M&E inputs, deliverable & risk tracking, stakeholder communication, survey & field support
Field Instruments	CTD, multi-parameter water meter, Algal Torch, Spectrophotometer, Atomic Absorption Spectrophotometer (AAS)
Languages	Bangla (Native/Fluent) | English (Fluent, working proficiency)

PROFESSIONAL EXPERIENCE

Project Coordinator
Jan 2026 – Present

Youth4Climate Mangrove Restoration and Monitoring Project • *Contractual*

- Coordinate data-driven mangrove restoration activities, including planning, field-validation preparation, stakeholder communication, documentation, M&E inputs, and progress tracking.
- Support GIS/RS-based site prioritization, restoration planning, and risk tracking using satellite data and Python/Google Earth Engine workflows.

Research Associate
Oct 2024 – Present

Bangladesh Maritime University • *Part-Time*

- Support climate, coastal, water-quality, and environmental research through literature review, data collection, analysis, documentation, and technical reporting.

- Prepare maps, figures, tables, presentations, and organized research outputs using GIS/RS, Python/R/SPSS, Excel, and visualization workflows.

Assistant Research Officer
Jun 2025 – Apr 2026

DSD Research Firm • Part-Time

- Supported proposal preparation, field coordination, questionnaire-based data collection, report writing, and deliverable management.
- Assisted with data cleaning, validation, summary table preparation, and evidence documentation for applied research assignments.

Data Analyst
Dec 2024 – May 2025

DSD Research Firm • Contractual

- Cleaned, structured, validated, and analyzed project datasets using Excel, Python, R, SPSS, Power BI, ArcGIS Pro, and QGIS.
- Prepared charts, dashboards, maps, tables, and interpretation note to support reports, presentations, manuscripts, and decision-making.

Researcher
Jan 2021 – Dec 2024

National Oceanographic and Maritime Institute (NOAMI) • Part-Time

- Assisted oceanographic and environmental studies through fieldwork, data collection, instrument support, sample documentation, dataset processing, and technical reporting.

SELECTED RESEARCH PROJECTS

- **Youth4Climate Mangrove Restoration and Monitoring Project —** Project coordination; GIS/RS site prioritization; ground-truthing; restoration planning; M&E and stakeholder coordination.
- **Climate-Induced Shifts in Potential Fishing Zones —** Fisheries, blue economy, and coastal livelihood resilience analysis using satellite-derived oceanographic data.
- **Existing and Potential Marine Protected Areas of Bangladesh —** Biodiversity conservation, area-based management, spatial assessment, and sustainable marine-management planning.
- **Sundarbans Mangrove Ecosystem Health Assessment —** Ecosystem condition, resilience, conservation concerns, and climate-vulnerability context.
- **Dhaka (Shitalakshya) River Ecosystem Health Assessment —** Water quality, ecological stress, and river-health dynamics for management decisions.
- **Marine Heatwaves and Coastal Bangladesh —** Thermal-stress patterns and ecological implications of climate-driven ocean stress.
- **Endangered Species Assessment Using eDNA —** Conservation-focused research using environmental DNA for aquatic species detection and monitoring.

PUBLICATIONS & MANUSCRIPTS

- Sowrav, S. F. F., et al., and Hasan, M. A. (2026). Water quality affects socioeconomic traits of climate-vulnerable coastal people: case study on Dacope and Mongla. Journal of Integrated Disaster Risk Management, 16(1). DOI: 10.5595/001c.158653.
- Hasan, Md. A., et al. (2026). Assessing the dynamics of water quality and implications on bio-chemical health of river ecosystem: A RS-based study of the Shitalakshya River, Dhaka. Bangladesh Maritime Journal, 10(1), 313–329. DOI: 10.70279/bmj-v2-1124.

- **Manuscripts in progress / under review:** potential fishing zone (PFZ) variability; Dhaka River water quality; Sundarbans mangrove restoration; OMZ distribution modelling; marine heatwave exposure; air quality assessment; marine spatial planning for Bangladesh's EEZ.

EDUCATION

Degree/Exam	Institution/Board	Passing Year	Result
M.Sc. in Oceanography	Bangladesh Maritime University	Expected Dec 2026	Thesis ongoing
B.Sc. in Oceanography	Bangladesh Maritime University	2025	CGPA 3.55/4.00
HSC (Science)	Dhaka Board	2019	GPA 4.67/5.00
SSC (Science)	Dhaka Board	2017	GPA 4.91/5.00

TRAINING & CERTIFICATIONS

- Climate Change, & Research Methodology — CCDB, Dec 2025
- Marine Megafauna and Fisheries Survey Methodology — WCS, 2025
- Workshop on Coastal & Regional Ocean Community Model (CROCO) — BMU, Aug 2025
- Earth Observation for Enhanced Fisheries MCS — UNESCO/IOC, Jul 2025
- Numerical Model & AI for Ocean Science (NMAI4OS) — NOAMI, 2024
- Radar Meteorology Applied Lecture Series — SAMA/ACARR-CUSAT, 2024 (Outstanding)
- SeaDAS for Optical Remote Sensing & Water Quality Monitoring — NASA ARSET, Feb 2024
- Satellite Meteorology Lecture Series — SAMA/BIT Mesra, 2023–2024
- Ocean Model for OFS Implementation — UNESCO/IOC & BMKG, Jul 2023
- Building Research Career for Early-Career Professionals & Students — Nirmol BD, 2023
- Lecture Series in Atmospheric & Climate Sciences — SAMA/SRM IST, Jan–Apr 2023
- Data Science & Machine Learning with Python — NOAMI & BCA, 2021

AWARDS & LEADERSHIP

Climate Change Scholarship—CCDB (2025)	Founding Team Head, BMU Ocean Club (2024-2025)
3rd Place Winner—NOAMI Ocean Olympiad (2025)	
Best Trainee—Radar Meteorology Lecture Series, SAMA (2024)	Advisor, BMU Ocean Club (2026-Present)
Best Trainee — Data Science with Python, NOAMI (2022)	

REFERENCES

Dr. S. M. Mustafizur Rahman — Assistant Professor, Bangladesh Maritime University | mustafizur.ocn@bmu.edu.bd | +8801401807894

Dr. Mohan Kumar Das — Executive Director, Hydro-Climate & Ocean Center (HCO) | mohan.smrc@gmail.com | +8801913630710

Sheikh Fahim Faysal Sowrav — Lecturer, Bangladesh Maritime University | sowrav.ocean@gmail.com | +8801986761741